Confidence-Driven Design Optimization Using Gaussian Process Metamodeling With Insufficient Data

Mingyang Li
Department of Mechanical Engineering-Engineering Mechanics,
Michigan Technological University,
Houghton, MI 49931
e-mail: mli7@mtu.edu

Zequn Wang[1]
Department of Mechanical Engineering-Engineering Mechanics,
Michigan Technological University,
Houghton, MI 49931
e-mail: zequnw@mtu.edu

To reduce the computational cost, surrogate models have been widely used to replace expensive simulations in design under uncertainty. However, most existing methods may introduce significant errors when the training data is limited. This paper presents a confidence-driven design optimization (CDDO) framework to manage surrogate model uncertainty for probabilistic design optimization. In this study, a confidence-based Gaussian process (GP) modeling technique is developed to handle the surrogate model uncertainty in system performance predictions by taking both the prediction mean and variance into account. With a target confidence level, the confidence-based GP models are used to reduce the probability of underestimating the probability of failure in reliability assessment. In addition, a new sensitivity analysis method is proposed to approximate the sensitivity of the reliability at the target confidence level with respect to design variables, and thus facilitate the CDDO framework. Three case studies are introduced to demonstrate the effectiveness of the proposed approach.
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1 Introduction

In reliability-based design optimization (RBDO) [1–5], reliability is defined as the probability that the system performs its intended function while considering various sources of uncertainties. By solving RBDO problems, optimum system designs that have the best compromise between cost and system reliability can be obtained accordingly. In the literature, sophisticated optimization strategies such as double-loop [6,7], decoupled-loop [8,9], and single-loop approaches [10,11] have been developed to improve the computational efficiency of RBDO. In the double-loop approaches, reliability analysis is performed for every design iteration, resulting in a nesting of two distinct levels of optimization. In contrast, single-loop approaches replace the probabilistic constraints with optimality conditions and solve the RBDO in a single loop procedure, while decoupled-loop methods transform the RBDO problem into a sequence of deterministic optimization problems. Besides the optimization strategies, performing an accurate yet efficient reliability analysis is of critical importance in RBDO as it involves repeated expensive simulations. Both analytical- and simulation-based approaches [12,13] have been developed to improve the performance of reliability analysis. Among analytical-based reliability analysis methods [14–16], the first-order reliability method transforms the random variables into a standard normal space (U-space) and approximates reliability by calculating the reliability index defined as the distance between the origin to the most probable point. Since the performance function is approximated by the first- or second-order Taylor expansion, the first-order reliability method and second-order reliability method may introduce significant errors when dealing with highly nonlinear problems. Compared with the analytical-based methods, simulation-based methods [17,18] are capable of providing more accurate reliability estimations. However, the direct Monte Carlo simulation (MCS) is computationally prohibitive in practical engineering applications.

As a technique of replacing expensive physics-based models by easy-to-evaluate mathematical functions, surrogate modeling such as radial basis functions [19–21], Gaussian process (GP) modeling (or Kriging modeling) [22,23], artificial neural networks [24–26], and polynomial response surface [27,28] have been extensively used to alleviate the computational burden in reliability analysis and design under uncertainty. The major concern of using surrogates in RBDO lies in that it still lacks rigorous methods for quantifying and managing the surrogate model uncertainty due to insufficient training data in reliability analysis. Different strategies have been proposed to manage the surrogate model uncertainty: (1) iteratively updating the surrogate model through adaptive sampling methods [29–34], and 2) using conservative surrogate modeling methods to provide safer response predictions [35–37]. By using the support vector machine for surrogate modeling, Basudhar and Missoum [38] developed an adaptive sampling scheme that the sample point with the highest probability of being misclassified will be selected to update the model. Lee and Jung [39] proposed a constraint boundary sampling technique to update the Kriging model by sequentially identifying sample points near the constraint boundary. Wang and Wang [40] proposed a maximum confidence enhancement-based sequential sampling scheme to quantify the accuracy of reliability estimations using cumulative confidence measure, then enhance the fidelity of surrogate models by identifying the most useful points iteratively. Zhao et al. [41] and Echard et al. [42] utilized Kriging prediction variance to select the best additional training samples for updating surrogate models. Chen et al. [43] proposed a local adaptive sampling method to efficiently update the Kriging model in RBDO, where a critical local adaptive sampling region of the design point is constructed to find new training samples. Adaptive sampling methods update the surrogate model by adding more training samples; however, the number of function evaluations is severely limited in most complex engineering applications due to the high computational costs, especially for high-dimensional problems. In such cases, surrogate model uncertainty is involved and large errors may be introduced in predicting the system performances. To handle the surrogate model uncertainty without adding extra training samples, conservative surrogate modeling techniques have been developed in recent years. As the most straightforward method,

[1]Corresponding author.

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safety margin approach biases the surrogate model predictions by adding an additive constant. However, it was reported [44] that there is no physical-based practice for choosing the constant. Viana et al. [45] employed the cross-validation method to provide a reasonable magnitude of the safety margin for a given level of conservativeness. Besides the safety margin method, Zhao et al. [46] employed the corrected Akaike information criterion to build a conservative Kriging model. Picheny et al. [47] utilized biased fitting to approximate the cumulative distribution function of random system responses, and thus overestimate the probability of failure and achieve conservative reliability approximations. Sjöstedt-de Luna and Young [48] employed the bootstrap method to construct conservative surrogate models by replacing the original Kriging prediction variance with the bootstrapping variance. Sensitivity analysis is often required to search for the optimal solution in RBDO. In surrogate model-based RBDO, the finite difference method is commonly used for approximating the sensitivity; nevertheless, the accuracy is highly affected by the step size and it lacks efficiency for high dimensional problems because reliability analysis must be performed for all perturbed design points. The first-order score function method [49] has been integrated with Monte Carlo simulation to efficiently perform sensitivity analysis. Though the first-order score function method does not incur any extra function evaluations, the approximated sensitivity is not smooth especially when the estimated reliability approaches zero or one, which may result in the increasing of design iterations and sometimes nonconvergence of RBDO.

To overcome the challenges, we propose an effective confidence-driven design optimization (CDDO) framework for surrogate-based design optimization under uncertainty. The merits of the proposed work are summarized as follows: (1) by efficiently handling the surrogate model uncertainty due to insufficient data, we propose a new confidence level-based reliability analysis approach to avoid the underestimation of probability of failure, (2) a new sensitivity approach is proposed for ensuring smooth optimization process, and (3) the confidence-driven design optimization framework will facilitate the decision making by balancing the trade-off between the risk due to lack of training data and the objective cost function. The rest of the paper is organized as follows: Section 2 briefly reviews the safety margin approach for conservative surrogate modeling and Sec. 3 details the proposed confidence-driven design optimization framework. Three case studies are solved to demonstrate the effectiveness of the proposed approach in Sec. 4.

2 Conservative Surrogate Modeling Using Safety Margin

Given a set of training points, a surrogate model $\hat{g}(\mathbf{x})$ can be constructed after evaluating the responses of all the training points. By adding a safety margin to the surrogate model, a conservative surrogate model $\hat{g}_c(\mathbf{x})$ can be obtained as

$$\hat{g}_c(\mathbf{x}) = \hat{g}(\mathbf{x}) - s \tag{1}$$

where $\mathbf{x}$ is a vector of system input parameters and s is the safety margin used to bias the predicted response from the original surrogate model $\hat{g}(\mathbf{x})$. The safety margin s is always a constant and its sign depends on the definition of the failure event. Assuming a failure event occurs if the system performance is less than zero, the safety margin will always be positive. As shown in Eq. (1), determining a suitable safety margin is of critical importance in compensating the surrogate model uncertainty due to the lack of data in reliability analysis.

In the literature, the prediction errors between the surrogate model and the origin performance function are calculated for computing the safety margin, given as

$$e(\mathbf{x}_t) = \hat{g}(\mathbf{x}_t) - g(\mathbf{x}_t) \tag{2}$$

where $\mathbf{x}_t$ is the vector of a large number of test samples within the design space. Accordingly, the cumulative distribution function (CDF) of the prediction errors F_e can be calculated and the fraction of the errors can be obtained by

$$c = 1 - F_e(0) \tag{3}$$

where c is referred to as the conservativeness level of the surrogate model. For a given conservativeness level, the safety margin can be estimated in terms of the cumulative distribution function F_e as

$$s = -F_e^{-1}(1 - c) \tag{4}$$

where F_e^{-1} is the inverse cumulative distribution function of the prediction errors. Obviously, the number of test samples determines the accuracy of the inverse CDF in Eq. (4) and affects the obtained safety margin. Therefore, obtaining an accurate estimation of the inverse CDF is extremely important in ensuring the performance of conservative estimations. By using a large number of test samples, an accurate safety margin for achieving a given conservativeness level can be obtained by Eq. (4). However, it is not applicable in practice due to the lack of training data. To alleviate the computational burden, Vianna et al. [45] employed the cross-validation method to approximate the prediction errors. For a surrogate model constructed with p training samples, cross-validation is a process of constructing new surrogate models without one of the p points and calculating the error respectively. After repeating the leave-one-out strategy for all the p samples, the corresponding cross-validation errors can be obtained and used as the prediction errors of the origin surrogate model.

3 Confidence-Driven Design Optimization

This section presents the confidence-driven design optimization framework for probabilistic design optimization. Subsection 3.1 introduces the Gaussian process modeling technique and the concept of confidence level. Subsection 3.2 details the reliability analysis using the confidence-based predictions and Monte Carlo simulation. A new sensitivity analysis method for CDDO is proposed in Subsec. 3.3, while Subsec. 3.4 summarizes the numerical procedure of the CDDO framework.

3.1 Confidence-Based Gaussian Process Modeling. In GP modeling, training sample points are involved in both approximating model parameters and predicting the response as

$$\hat{g}_{gp}(\mathbf{x}) \sim GP\left(\mathbf{h}(\mathbf{x})\boldsymbol{\beta}, \sigma^2\mathbf{R}\right) \tag{5}$$

where $\hat{g}_{gp}(\mathbf{x})$ is the approximation of the expensive performance function $G(\mathbf{x})$ at the point $\mathbf{x}$; $\mathbf{h}(\mathbf{x})$ is a row vector of regression functions; $\boldsymbol{\beta}$ is a column vector of the coefficients; $\sigma^2\mathbf{R}$ is the covariance matrix; σ is an unknown hyperparameter; $\mathbf{R}$ represents the correlation matrix. The terms $\mathbf{h}(\mathbf{x})$ and $\boldsymbol{\beta}$ are treated as unit vectors as they are sufficient for most engineering applications as introduced by the studies [50,51]. Given a training data set $\mathbf{D} = [\mathbf{X}, \mathbf{G}]$, where $\mathbf{X} = [\mathbf{x_1}, \mathbf{x_2}, \ldots, \mathbf{x_n}]$ represents n input points and $\mathbf{G}$ represents the corresponding performance $[G(\mathbf{x_1}), G(\mathbf{x_2}), \ldots, G(\mathbf{x_n})]$, the (i, j) entry of the correlation matrix can be expressed as

$$\mathbf{R}_{(i,j)} = \exp\left[-\sum_{q=1}^{k} a_q |x_{i,q} - x_{j,q}|^{b_q}\right] \tag{6}$$

where k is the dimension of the input $\mathbf{x}$, a_q and b_q $(q = 1, 2, \ldots, k)$ are hyperparameters of the GP model. Based on the training data set $\mathbf{D}$, these hyperparameters can be obtained through maximum likelihood estimation. At any given point $\mathbf{x}'$, the GP model is

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capable of providing the response prediction as a normal distribution, expressed as

$$\hat{g}_{gp}(\mathbf{x}') \sim N\left(\mu_{gp}(\mathbf{x}'), v_{gp}(\mathbf{x}')\right) \tag{7}$$

where the prediction mean μ_{gp} and prediction variance v_{gp} are given as

$$\mu_{gp}(\mathbf{x}') = \mathbf{h}(\mathbf{x})\boldsymbol{\beta} + \mathbf{r}^{\mathrm{T}}\mathbf{R}^{-1}(\mathbf{G} - \mathbf{H}\boldsymbol{\beta}) \tag{8}$$

$$v_{gp}(\mathbf{x}') = \sigma^2\Big\{1 - \mathbf{r}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{r} + \left[\mathbf{h}^{\mathrm{T}}(\mathbf{x}) - \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{r}\right]^{\mathrm{T}}(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H})^{-1} \left[\mathbf{h}^{\mathrm{T}}(\mathbf{x}) - \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{r}\right]\Big\} \tag{9}$$

where $\mathbf{r}$ is the correlation vector between the input $\mathbf{x}'$ and all training sample points $\mathbf{X}$, and $\mathbf{H} = [\mathbf{h}(\mathbf{x}_1), \mathbf{h}(\mathbf{x}_2), \ldots, \mathbf{h}(\mathbf{x}_n)]$. Due to the lack of training data, surrogate model uncertainty is involved in predicting the system performance function $G(\mathbf{x})$ using GP models.

Assuming a failure event occurs if the system performance is less than zero, for any given input $\mathbf{x}$', there are four potential scenarios as shown in Fig. 1, and two types of classification errors can be introduced due to the surrogate model uncertainty as illustrated in Figs. 1(*a*) and 1(*c*), termed as type I and II classification errors, respectively. The type I classification error occurs when the input point is identified as a failure but its true response is greater than zero. Similarly, the type II classification error occurs when the input point is identified as safe; however, it actually is a failure point. In reliability analysis, type II classification errors may lead to the underestimation of the probability of failure and therefore overestimate the reliability. Thus, the surrogate model uncertainty needs to be taken into consideration to avoid the type II classification errors. In this work, a confidence-based GP modeling technique is developed to rigorously account for surrogate model uncertainty in predicting system performance. As shown in Eq. (7), the response prediction for a given point $\mathbf{x}'$ is treated as a normally distributed random variable, denoted as $\hat{g}_{gp}(\mathbf{x}') \sim N(\mu_{gp}(\mathbf{x}'), v_{gp}(\mathbf{x}'))$. Therefore, a confidence level is defined based on this normal distribution, expressed as

$$\mathrm{CL} = \Phi\left(\frac{\hat{g}_{gp}(\mathbf{x}')|_{\mathrm{CL}} - \mu_{gp}(\mathbf{x}')}{\sqrt{v_{gp}(\mathbf{x}')}}\right) \tag{10}$$

where $\hat{g}_{gp}(\mathbf{x}')|_{\mathrm{CL}}$ represents the percentile value of the normally distributed response prediction. At a given confidence level, a confidence-based Gaussian process model can be constructed based on the first and second moments of the prediction from the GP model, expressed as

$$\hat{g}_{gp}(\mathbf{x}')|_{\mathrm{CL}} = \mu_{gp}(\mathbf{x}') - \Phi^{-1}(\mathrm{CL})\sqrt{v_{gp}(\mathbf{x}')} \tag{11}$$

where $\Phi^{-1}(.)$ is the inverse standard normal cumulative distribution function. In Fig. 1, only scenario (c) introduces type II classification errors. For scenario (c), the probability of type II classification error is less than (1 - CL), and it decreases to zero if the confidence level is greater than CLc that corresponds to zero confidence-based prediction. As shown in Fig. 2, the confidence-based prediction is monotonically decreasing with the value of CL and reaches its minimum at CL = 1. By adopting a certain confidence level CL for the GP prediction, the probability of type II classification errors will be less than (1-CL). By extending the confidence level concept in reliability analysis, we can mitigate the risk of underestimating the probability of failure.

3.2 Reliability Analysis With a User-Defined Confidence Level. For a performance function $G(\mathbf{x})$, the reliability analysis involves the calculation of the probability of failure, denoted by P_f, which is generally expressed as

$$P_f = \Pr(G(\mathbf{x}) < 0) = \int \ldots \int_{G(\mathbf{x})<0} f_x(\mathbf{x})d\mathbf{x} \tag{12}$$

where $\mathbf{x}$ is the vector of random variables and $f_x(\mathbf{x})$ is the corresponding joint probability density function. Let Ω_f

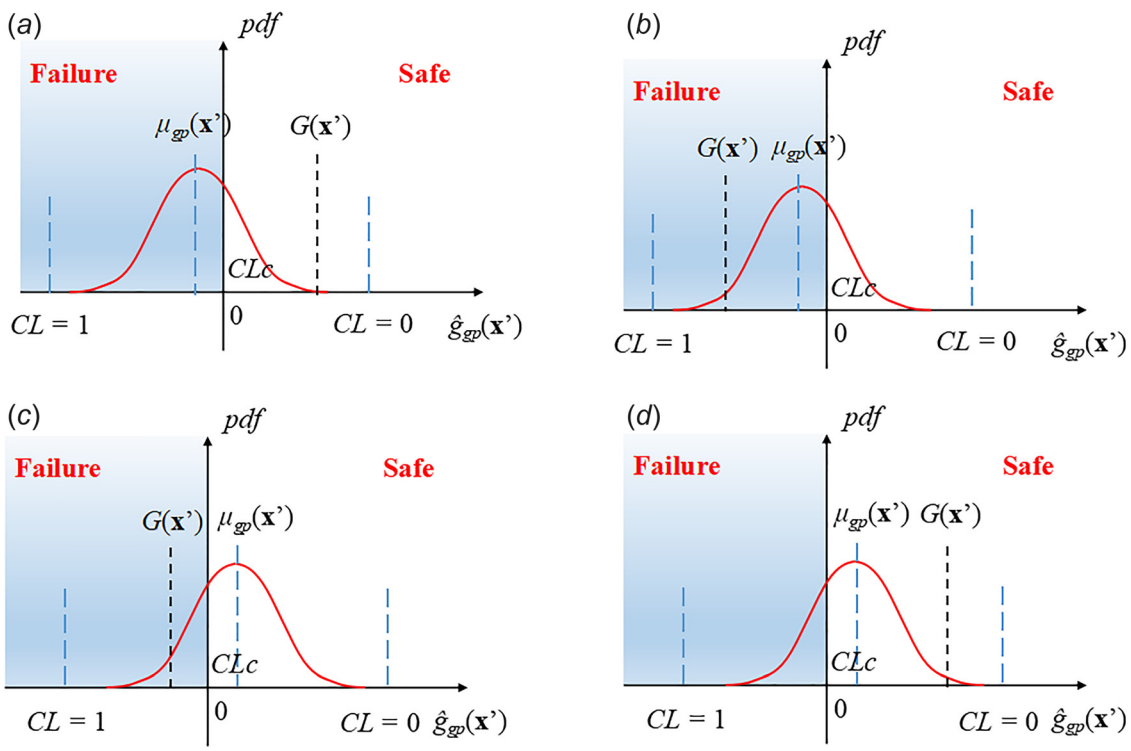


Fig. 1 Four scenarios, (*a*) and (*c*) represent type I and II classification errors, respectively

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represents the failure region, the probability of failure can be rewritten as

$$P_f = \Pr(\mathbf{x} \in \Omega_f) = \int_{\Omega} I_f(\mathbf{x}) f_x(\mathbf{x}) d\mathbf{x} = E\left[I_f(\mathbf{x})\right] \tag{13}$$

where $E[.]$ is the expectation operator and $I_f(\mathbf{x})$ is a failure indicator function. Sampling methods can be directly used for reliability analysis based on Eq. (13), and the Gaussian process modeling method is widely used to reduce the computational costs. In most of the existing GP model-based methods, the prediction mean is employed to identify potential failures based on the indicator function. When the surrogate model uncertainty is involved due to the lack of data, type II classification errors may appear and significant errors can be introduced in reliability analysis as illustrated in Sec. 3.1. By adopting the confidence-based prediction, the probability of failure can be approximated with the consideration of surrogate model uncertainty as

$$\hat{P}_f = \Pr\left[\left(\mu_{gp}(\mathbf{x}) - \Phi^{-1}(CL)\sqrt{v_{gp}(\mathbf{x})}\right) < 0\right] \tag{14}$$

Based on Eq. (14), the Monte Carlo simulation method can be directly utilized for reliability analysis with a target confidence level. For a performance function $G(\mathbf{x})$, a Gaussian process model is first constructed as introduced in Subsec. 3.1, then MCS generates N random samples according to the randomness of the input $\mathbf{x}$, denoted as $\mathbf{X}_m = [\mathbf{x}_{m,1}, \ldots, \mathbf{x}_{m,i}, \ldots, \mathbf{x}_{m,N}]$. The response of each MCS sample is then estimated by the GP model, and follows a normal distribution with prediction mean and variance. At a given confidence level, the confidence-based predictions $\hat{g}_{gp}(\mathbf{X_m})|_{\mathrm{CL}}$ with the consideration of surrogate model uncertainty can be calculated by using Eq. (11). To compensate for the type II classification errors, instead of using the prediction mean, the confidence-based predictions are used to classify the sample as failure or safe by an indicator function, given as

$$I_f(\mathbf{x}_{m,i}) = \begin{cases} 1, \hat{\mathrm{g}}_{gp}(\mathbf{x}_{m,i})|_{\mathrm{CL}} < 0 \text{ failure} \\ 0, \hat{\mathrm{g}}_{gp}(\mathbf{x}_{m,i})|_{\mathrm{CL}} \geq 0 \text{ safe} \end{cases} \tag{15}$$

It is worth noting that the difference between confidence-based prediction and prediction mean for each MCS sample is different because the prediction means and variances vary with the position of the MCS samples. After classifying all the MCS samples, the reliability can be simply approximated based on the ratio of the number of failure samples to the total number N, expressed as

$$\hat{R}_{\mathrm{CL}} = 1 - \hat{P}_f = 1 - \frac{1}{N}\sum_{i=1}^{N} I_f(\mathbf{x}_{m,i}) \tag{16}$$

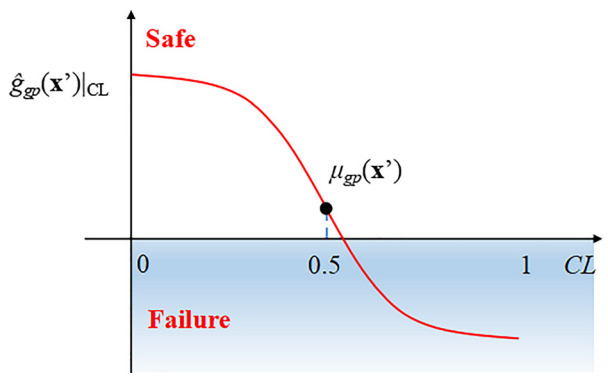


Fig. 2 Confidence-based predictions versus confidence level

In the proposed confidence-based reliability analysis approach, the confidence-based prediction is equal to the prediction mean when the confidence level is 0.5 and the term $\Phi^{-1}(\mathrm{CL})$ is always a negative value when CL is within the range [0.5, 1]. With a confidence level greater than 0.5, the confidence-based prediction of each MCS sample is smaller than the prediction mean. As a result, more samples will be identified as failure points in Eq. (16) and we are more likely to compensate for the type II classification errors in reliability analysis. Since the confidence-based prediction decreases with increasing confidence level, it is found that the estimated probability of failure is always monotonically increasing at the value of CL. Accordingly, the confidence level CL is treated as a quantitative measure to indicate the probability of overestimating the reliability. When classifying the MCS samples, a certain confidence level is adopted for each GP prediction. Note the probability of having the type II classification error for a single realization is less than (1-CL). By averaging all the N realizations, the probability of underestimating the probability of failure will be less than (1-CL) if the confidence-based predictions are used to approximate the reliability. For a reliability estimation $\hat{R}_{CL}$, a transformed reliability index β_C can always be obtained as

$$\beta_C = \Phi_t^{-1}\left(\hat{R}_{CL}\right) \tag{17}$$

where $\Phi_t^{-1}(.)$ is an inverse cumulative distribution function of a truncated normal distribution.

• *A two-dimensional Mathematical Example*. A two-dimensional example is used to test the performance of the proposed reliability analysis method, expressed as

$$G(x_1, x_2) = x_1^3 x_2 + 4x_2 - x_1^4 + 4x_1^2 - 8 \tag{18}$$

where the inputs x_1 and x_2 are two normally distributed random variables as $x_1 \sim N(u_1, 0.05)$ and $x_2 \sim N(u_2, 0.05)$, respectively. Within the boundary of [0, 3] for both x_1 and x_2, Latin hypercube sampling scheme is employed to generate seven samples for constructing a Gaussian process model. Figure 3 shows the contours of the estimated limit state function (LSF) and origin LSF, where the seven training points are depicted as black nodes. Obviously, the constructed GP model cannot provide accurate response predictions due to the surrogate model uncertainty. MCS with 10^6 samples is used for reliability assessment at six different design points, shown as the stars in Fig. 3. The results of reliability analysis using direct MCS, conventional method (CL = 0.5), and the

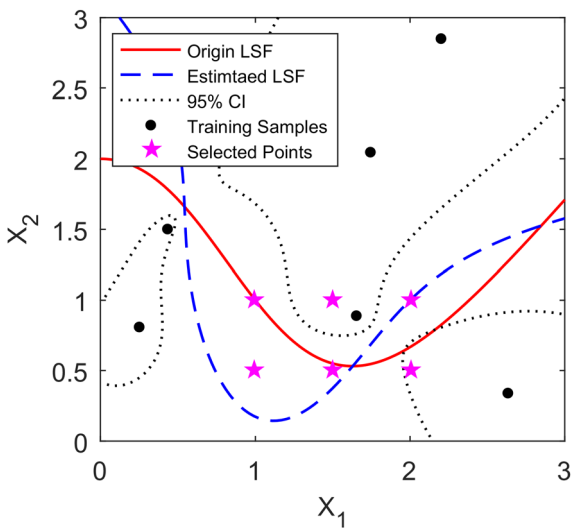


Fig. 3 Estimated versus origin LSF

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proposed approach at different CLs are summarized in Table 1. Conventional method overestimates the reliability since the surrogate model uncertainty is completely ignored. On the contrary, the proposed approach is capable of providing reliable reliability estimations by compensating the type II classification errors using the confidence-based predictions. At the point $\mathbf{x} = [1.5, 1]$, Fig. 4 shows the relationship between the estimated probability of failure and confidence level and Fig. 5 shows the estimated PDF of random outputs at different CLs.

3.3 Sensitivity Analysis in Confidence-Driven Design Optimization. In this section, we propose a new sensitivity analysis method for the confidence-driven design optimization. For a given confidence level, the reliability of a design point can be estimated based on the confidence-based predictions and MCS method as introduced in Subsec. 3.2. The reliability can be approximated according to the PDF of the predicted system response, expressed as

$$\hat{R}_{CL} = \Phi_t(\beta_C) = \Pr\left(\hat{g}_{gp}(\mathbf{x})|_{CL} \geq 0\right) = \int_0^{+\infty} f\left(\hat{g}_{gp}(\mathbf{x})|_{CL}\right) d\left(\hat{g}_{gp}(\mathbf{x})|_{CL}\right) \approx \Phi\left(\frac{\mu_C}{\sqrt{v_C}}\right) \tag{19}$$

where $f(\hat{g}_{gp}(\mathbf{x})|_{\mathrm{CL}})$ is the probability density function of the predicted system response at the given confidence level, $\Phi(.)$ is the standard normal cumulative distribution function, μ_C and v_C are the mean and variance of the confidence-based predictions $\hat{g}_{gp}(\mathbf{X_m})|_{\mathrm{CL}}$ of all the MCS samples, respectively. In Eq. (19), it shows that the transformed reliability index β_C can be estimated by the ratio of the mean to the standard deviation of the confidence-based predictions. Therefore, the sensitivity of transformed reliability index β_C can be approximated as

$$\frac{\partial \beta_C}{\partial \mathbf{d}} = \frac{\partial\left[\Phi_t^{-1}(\hat{R})\right]}{\partial \mathbf{d}} \approx \frac{\partial\left(\frac{\mu_C}{\sqrt{v_C}}\right)}{\partial \mathbf{d}} = \frac{1}{\sqrt{v_C}}\frac{\partial \mu_C}{\partial \mathbf{d}} - \frac{\mu_t}{2\sqrt{v_C}^3}\frac{\partial v_C}{\partial \mathbf{d}} \tag{20}$$

where the mean of the conservative response μ_C can be calculated as

$$\mu_C = \frac{1}{N}\sum_{i=1}^{N} \hat{g}_{gp}(\mathbf{x}_{m,i})|_{\mathrm{CL}} = \frac{1}{N}\sum_{i=1}^{N}\left[\mu_i - \Phi^{-1}(\mathrm{CL})\sqrt{v_i}\right] \tag{21}$$

and the variance v_C can be obtained as

$$v_C = \frac{1}{N}\sum_{i=1}^{N}\left[\mu_i - \Phi^{-1}(\mathrm{CL})\sqrt{v_i}\right]^2 - \mu_C^2 \tag{22}$$

where μ_i represents the prediction mean of the ith MCS sample $\mu_{gp}(\mathbf{x}_{\mathrm{m},i})$ and v_i represents the prediction variance $v_{gp}(\mathbf{x}_{\mathrm{m},i})$. In Eq. (20), the gradient of the mean μ_C and variance v_C can be obtained by taking the first-order derivative of Eqs. (21) and (22), given as

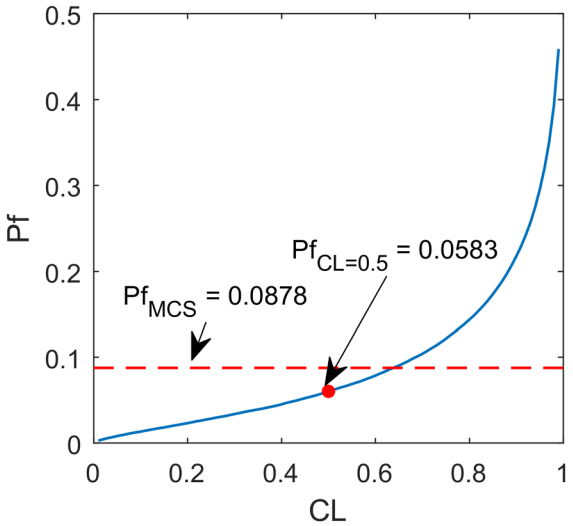


Fig. 4 Estimated probability of failure increases with higher CL (At x = [1.5, 1])

$$\frac{\partial \mu_C}{\partial \mathbf{d}} = \frac{1}{N}\sum_{i}^{N}\frac{\partial u_i}{\partial \mathbf{d}} - \Phi^{-1}(\mathrm{CL})\frac{1}{N}\sum_{i}^{N}\frac{\partial v_i}{\partial \mathbf{d}}\frac{1}{2\sqrt{v_i}} \tag{23}$$

$$\frac{\partial v_C}{\partial \mathbf{d}} = \frac{1}{N}\sum_{i}^{N}\left[2\left(u_i - \Phi^{-1}(\mathrm{CL})\sqrt{v_i}\right)\left(\frac{\partial u_i}{\partial \mathbf{d}} - \Phi^{-1}(\mathrm{CL})\frac{\partial v_i}{\partial \mathbf{d}}\frac{1}{2\sqrt{v_i}}\right)\right] - 2\mu_C\frac{\partial \mu_C}{\partial \mathbf{d}} \tag{24}$$

It is observed that Eqs. (23) and (24) are functions of the first-order derivative of the prediction mean μ_i and variance v_i for each MCS sample. In the Gaussian process model, the derivative of prediction mean and variance with respect to the input $\mathbf{x}$ can be analytically obtained according to Eqs. (8) and (9), expressed as

$$\frac{\partial \mu_i}{\partial \mathbf{x}} = \mathbf{J}_h\boldsymbol{\beta} + \mathbf{J}_r\mathbf{R}^{-1}(\mathbf{G} - \mathbf{H}\boldsymbol{\beta}) \tag{25}$$

$$\frac{\partial v_i}{\partial \mathbf{x}} = 2\sigma^2\left[-\mathbf{R}^{-1}\mathbf{J}_r + \left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\left(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{J}_r - \mathbf{J}_h\right)\right] \tag{26}$$

where σ^2 is the hyperparameter of the GP model, $\mathbf{J}_{\mathrm{h}}$ and $\mathbf{J}_{\mathrm{r}}$ are the Jacobian of $\mathbf{h}(\mathbf{x})$ and $\mathbf{r}(\mathbf{x})$, respectively. As introduced in Subsec. 3.1, $\mathbf{h}(\mathbf{x})$ is a unit vector and $\mathbf{r}(\mathbf{x})$ represents the correlation vector between the point $\mathbf{x}$ and the training sample points. Therefore, Eqs. (23) and (24) can be easily evaluated after calculating Eqs.

Table 1 Estimated reliability at different CL

	Confidence level					
Point	0.6	0.7	0.8	0.9	MCS	Conventional (CL= 0.5)
[1, 0.5]	0.5089	0.3159	0.1692	0.0677	0.0852	0.7414
[1, 1]	0.8885	0.8007	0.6531	0.4149	0.4864	0.9350
[1.5, 0.5]	0.6021	0.3454	0.2497	0.1662	0.3343	0.5997
[1.5, 1]	0.9230	0.8934	0.8576	0.7856	0.9122	0.9417
[2, 0.5]	0.0818	0.0628	0.0509	0.0385	0.2434	0.1047
[2, 1]	0.4681	0.4138	0.3560	0.2908	0.8406	0.5234

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(25) and (26) for each MCS sample. After that, the obtained gradient terms are further used in Eq. (20) to approximate the sensitivity of the transformed reliability index β_C with respect to design variables.

3.4 Confidence-Driven Design Optimization Framework. In confidence-driven design optimization, a predefined confidence level is set as a target and the reliability analysis is performed at this specified CL. The design objective is to minimize the cost function while ensuring the reliability assessment at the target confidence level satisfies the reliability requirement. Therefore, the proposed CDDO framework is formulated as

$$\begin{aligned} &\text{minimize}: \text{Cost}(\mathbf{X}, \mathbf{d}) \\ &\text{subject to}: \hat{R}^i_{\text{CL}} = \Pr\left(\hat{g}_{gp}(\mathbf{X}, \mathbf{d})|_{\text{CL}} \geq 0\right) \geq R^i_t, i = 1, 2, \ldots, nc \\ &\qquad \mathbf{d}^{\mathbf{L}} \leq \mathbf{d} \leq \mathbf{d}^{\mathbf{U}}, \quad \mathbf{d} \in R^{nd}, \ \mathbf{X} \in R^{nr}, \text{and CL} = \text{CL}_t \end{aligned} \tag{27}$$

where Cost($\mathbf{X}$, $\mathbf{d}$) is the objective function; $\hat{g}_{gp}(\mathbf{X}, \mathbf{d})|_{\text{CL}}$ is the confidence-based Gaussian process model of the ith limit state function $G_i(\mathbf{X}, \mathbf{d})$; $\hat{R}^i_{CL}$ is the corresponding estimated reliability and R_t^i is the target reliability; CL_t is the target confidence level; $\mathbf{d}$ is a vector of design variables and $\mathbf{X}$ is a vector of random variables; $\mathbf{d}_L$ and $\mathbf{d}_U$ are the lower and upper boundaries of the design variables; nc, nd, and nr are the numbers of constraints, design variables, and random variables, respectively. By using the transformed reliability index β_C, the CDDO formulation can be rewritten as

$$\begin{aligned} &\text{minimize}: \text{Cost}(\mathbf{X}, \mathbf{d}) \\ &\text{subject to}: \beta^i_C \geq \beta^i_t = \Phi^{-1}_t\left(R^i_t\right) \\ &\qquad \beta^i_C = \Phi^{-1}_t\left[\Pr\left(\hat{g}_{gp}(\mathbf{X}, \mathbf{d})|_{\text{CL}} \geq 0\right)\right], i = 1, 2, \ldots, nc \\ &\qquad \mathbf{d}^{\mathbf{L}} \leq \mathbf{d} \leq \mathbf{d}^{\mathbf{U}}, \quad \mathbf{d} \in R^{nd}, \ \mathbf{X} \in R^{nr}, \text{and CL} = \text{CL}_t \end{aligned} \tag{28}$$

where ${\beta_t}^i$ is the target reliability index.

The CDDO framework is summarized in Fig. 6 and Algorithm 1. Within the design domain, a set of n training samples $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$ are first generated by Latin hypercube sampling scheme. For a design problem with k-dimensions, each training sample is a vector of k variables. After evaluating the performance functions for all the training samples, nc initial training data sets can be obtained, denoted as $\mathbf{D}j = [\mathbf{X}, \mathbf{G}_j]$, where $j = 1, 2, \ldots, nc$ represents the jth limit state function. As a result, nc GP models can be constructed for representing the corresponding limit state functions. Starting with an initial design point, the confidence-driven design optimization framework will be performed iteratively to search for optimal solutions. In each design iteration, the proposed reliability analysis method at the target confidence level is employed to approximate the conservative reliability with the consideration of surrogate model uncertainty. N Monte Carlo samples are first generated according to the randomness of the current design point, denoted as $\mathbf{X_m} = [\mathbf{x}_{\text{m},1}, \mathbf{x}_{\text{m},2}, \ldots, \mathbf{x}_{\text{m},N}]$. The responses of the MCS samples are then predicted by the constructed GP models. At the target confidence level, the confidence-based predictions $\hat{g}_{gp}(\mathbf{X_m})|_{\text{CL}}$ can be calculated based on the prediction means and variances as shown in Eq. (11). By classifying the MCS samples using the indicator function in Eq. (15), all the failure samples will be counted for approximating the reliability. A transformed reliability index β_C is then obtained through the inverse CDF of a truncated normal distribution. By using the new formulation of sensitivity analysis, the sensitivity of β_C with respect to design variables is approximated based on the gradient of the mean μ_C and variance v_C of the confidence-based predictions $\hat{g}_{gp}(\mathbf{X_m})|_{\text{CL}}$. Once the sequential quadratic programming optimizer receives the transformed reliability index and the values of sensitivity, a new design point will be generated accordingly and the CDDO process will be repeated iteratively until an optimum design is obtained.

Algorithm 1: Procedure of CDDO framework

Initialization:
Initial design for optimization, $\mathbf{d}_0$
Target confidence level, CL_t
Size of the Monte Carlo simulations N
Target reliability index β_t
LHS samples $\mathbf{X}$ and corresponding date set $\mathbf{D}j$ ($j = 1, 2, \ldots, nc$)
GP models $\hat{g}^j_{gp}$ based on $\mathbf{D}j$ ($j = 1, 2, \ldots, nc$)
(SQP for optimization)
$\mathbf{d} = \mathbf{d}_0$;
iteration = 0;
for $j = 1$ to nc **do**
  for $i = 1$ to N **do**

$$\hat{g}^j_{gp}(\mathbf{x}_{m,i})|_{CL} = \mu^j_{gp}(\mathbf{x}_{m,i}) - \Phi^{-1}(CL)\sqrt{v^j_{gp}(\mathbf{x}_{m,i})}$$

  If $\hat{g}^j_{gp}(\mathbf{x}_{m,i})|_{CL} < 0$ **then**
  $N^j_f = N^j_f + 1$;
  end if
  gradient for each MCS sample $\dfrac{\partial \mu^j_i}{\partial \mathbf{x}}$ and $\dfrac{\partial v^j_i}{\partial \mathbf{x}}$
  end for
  Reliability analysis

$$P^j_f = \frac{N^j_f}{N}$$

  Sensitivity analysis:

$$\beta^j_c = \Phi^{-1}\left(1 - P^j_f\right)$$

$$\mu^j_C = \frac{1}{N}\sum_{i=1}^{N} \hat{g}^j_{gp}(\mathbf{x}_{m,i})|_{CL} = \frac{1}{N}\sum_{i=1}^{N}\left[\mu^j_i - \Phi^{-1}(CL)\sqrt{v^j_i}\right]$$

$$v^j_C = \frac{1}{N}\sum_{i=1}^{N}\left[\mu^j_i - \Phi^{-1}(CL)\sqrt{v^j_i}\right]^2 - \left(\mu^j_C\right)^2$$

$$\frac{\partial \mu^j_C}{\partial \mathbf{d}} = \frac{1}{N}\sum_{i}^{N}\frac{\partial u^j_i}{\partial \mathbf{d}} - \Phi^{-1}(CL)\frac{1}{N}\sum_{i}^{N}\frac{\partial v^j_i}{\partial \mathbf{d}}\frac{1}{2\sqrt{v^j_i}}$$

$$\frac{\partial v^j_C}{\partial \mathbf{d}} = \frac{1}{N}\sum_{i}^{N}\left[2\left(u^j_i - \Phi^{-1}(CL)\sqrt{v^j_i}\right)\left(\frac{\partial u^j_i}{\partial \mathbf{d}} - \Phi^{-1}(CL)\frac{\partial v^j_i}{\partial \mathbf{d}}\frac{1}{2\sqrt{v^j_i}}\right)\right] - 2\mu^j_C\frac{\partial \mu^j_C}{\partial \mathbf{d}}$$

$$\frac{\partial \beta^j_c}{\partial \mathbf{d}} = \frac{1}{\sqrt{v^j_C}}\frac{\partial \mu^j_C}{\partial \mathbf{d}} - \frac{\mu_C}{2\sqrt{v^j_C}^3}\frac{\partial v^j_C}{\partial \mathbf{d}}$$

end for
Check convergence of the optimization algorithm
If not converged
Update the design point $\mathbf{d}$
Iteration=iteration + 1;
If converged;
Optimum is achieved $\mathbf{d}_{\text{opt}} = \mathbf{d}$

4 Case Studies

In this section, three examples will be introduced to demonstrate the effectiveness of the proposed confidence-driven design optimization approach.

4.1 Case Study I: A Mathematical Design Problem. The first case study considers a benchmark mathematical design problem, in which two design variables $\mathbf{d} = [X_1, X_2]$ are both normally distributed with a variance 0.3464^2. The CDDO problem is formulated as

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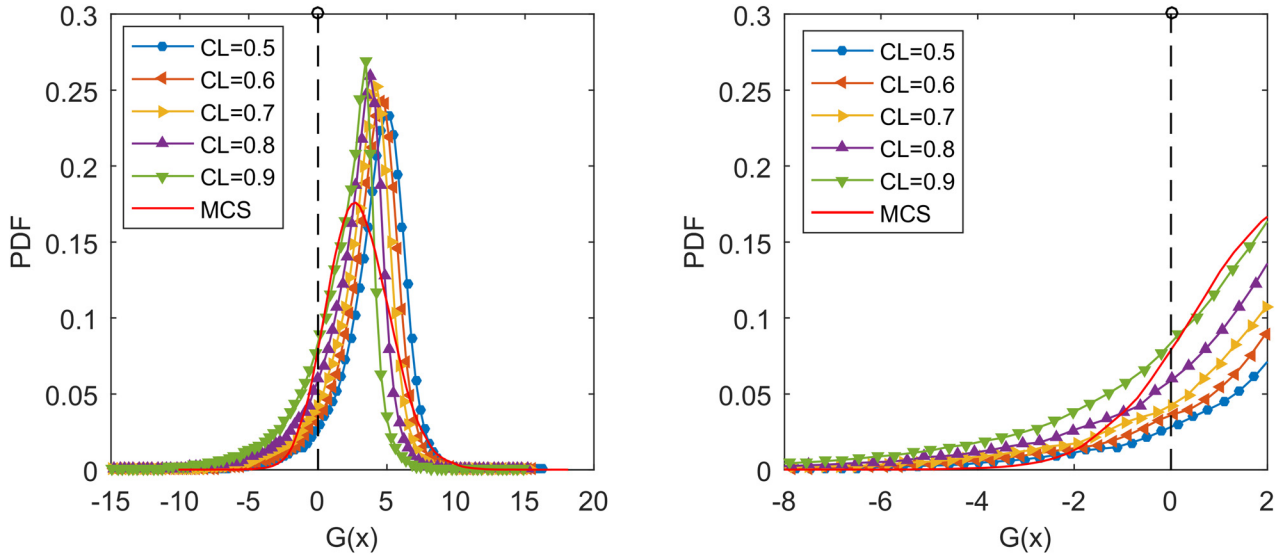


Fig. 5 Estimation for random outputs at different CL (At x = [1.5, 1])

$$
\begin{aligned}
&\text{minimize: Cost} = 10 - X_1 + X_2 \\
&\text{subject to: } \beta_C^i \geq \beta_t^i; \beta_C^i = \Phi_t^{-1}\left[\Pr\left(\hat{g}_{gp}^i(\mathbf{d})|_{\text{CL}} \geq 0\right)\right], i = 1 \sim 3 \\
&\quad \mathbf{d} = [X_1, X_2], \text{CL} = \text{CL}_t \\
&\quad 0 \leq X_1 \& X_2 \leq 10 \\
&\quad \text{where } G_1 = \frac{X_1^2 X_2}{20} - 1 \\
&G_2 = \frac{(X_1 + X_2 - 5)^2}{30} + \frac{(X_1 - X_2 - 12)^2}{120} - 1 \\
&G_3 = -0.05X_1^2 - X_1 - X_2 + 14
\end{aligned} \tag{29}
$$

In this study, the target reliability level is set to 0.985 for all the three constraints, while the target confidence level is set to 0.8. As outlined in Subsec. 3.4, the first step of employing the CDDO framework is to construct surrogate Gaussian process models for each limit state function. Within the design domain, six training samples are generated by Latin hypercube sampling and evaluated for the performances. As a result, three GP models can be constructed based on the obtained data sets $\mathbf{D}_i = [\mathbf{X}, \mathbf{G}_i]$, where $i = 1$, 2, 3. The CDDO framework is performed iteratively by starting from an initial design $\mathbf{d_0} = [5, 5]$. In each design iteration, 10^5 MCS samples are generated according to the randomness of the current design point for reliability analysis at the target confidence level. After achieving the transformed reliability index, the MCS samples are further utilized for sensitivity analysis as introduced in Subsec. 3.3. Eventually, an optimum design $\mathbf{d_{opt}} = [6.1813, 4.0147]$ is obtained after nine iterations. The iterative design history of design points, estimated reliabilities, and cost function values are listed in Table 2. Figure 7 shows the contours of the origin and estimated limit state functions using the prediction means. The six training samples are marked as black nodes. It is observed that the large prediction variances caused by the surrogate model uncertainty lead to wide 95% confidence intervals, which demonstrates that GP models constructed by six training samples cannot accurately represent the origin limit state functions.

The iterative design history for design variables X_1 and X_2 is shown in Fig. 8, which indicates that the optimum design converges after the fifth design iteration. Figure 9 shows the estimated reliabilities of three limit state functions during the optimization process. To test the performance of the proposed work, this mathematical design optimization problem is solved at different target confidence levels and the results are shown in Table 3. For comparison purpose, conventional surrogate model-based methods, safety margin method using 50 test samples, denoted as SM(TS), and safety margin method using cross-validation denoted as SM(CV) are employed for the same design problem, while the conservativeness level for the safety margin methods is set to 90%. Moreover, direct MCS method is used to provide an optimum design, which is treated as a reference. The optimum solutions obtained by using the four aforementioned methods are summarized in Table 3, where the reliabilities at these optimum designs are verified by MCS using 10^6 samples. Generally, reliability-based design optimization is computationally expensive as the expensive simulations are repeatedly evaluated for approximating the probability of failure. Therefore, the number of function evaluations $\#F$ has been treated as a measure of computational efficiency. In Table 3, the last column shows the number of function evaluations required by different methods. A contour plot of the comparison of different optimum designs is depicted in

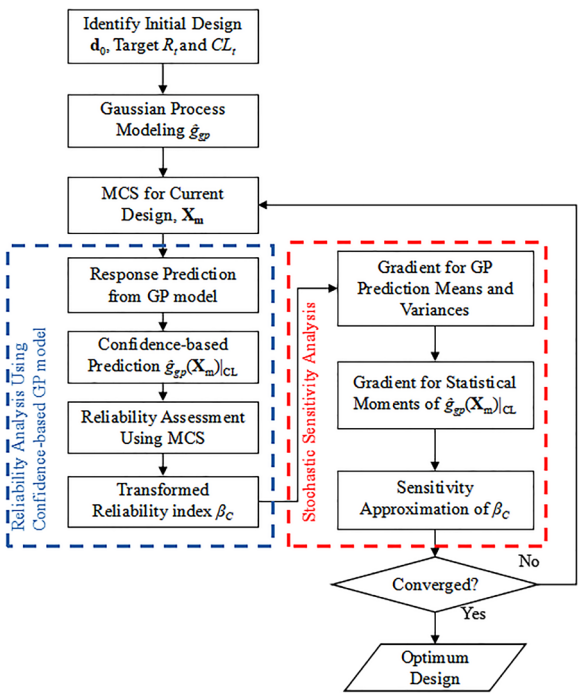


Fig. 6 Flowchart of confidence-driven design optimization

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Table 2 Design history for case study I

Iteration	Design point	$\hat{\mathbf{R}}_1$	$\hat{\mathbf{R}}_2$	$\hat{\mathbf{R}}_3$	Cost
1	[5.0000, 5.0000]	0.8490	0.9998	0.9996	10.0000
2	[5.6410, 4.6754]	0.9823	0.9989	0.9914	9.0344
3	[6.1454, 3.9791]	0.9928	0.9795	0.9898	7.8337
4	[6.1160, 4.0621]	0.9932	0.9858	0.9876	7.9461
5	[6.1079, 4.0824]	0.9933	0.9867	0.9871	7.9744
6	[6.1436, 4.0454]	0.9943	0.9857	0.9863	7.9017
7	[6.1715, 4.0228]	0.9950	0.9852	0.9854	7.8513
8	[6.1639, 4.0310]	0.9949	0.9854	0.9857	7.8671
9	[6.1813, 4.0147]	0.9954	0.9852	0.9849	7.8334

Fig. 10. Due to the ignorance of the surrogate model uncertainty, conventional methods provide an optimum design located in the failure region, where the reliabilities of the three constraints are calculated as 0.2375, 0.0045, and 0.8845, respectively. Safety margin using cross-validation method failed to offer a feasible solution because the estimated margin is too large when the training data is limited. On the contrary, SM(TS) method successfully obtains an acceptable optimum at 90% conservative level. However, the magnitude of the safety margins is approximated by comparing the difference between origin and predicted responses at the 50 test samples, leading to 150 more function evaluations than other approaches. With the consideration of surrogate model uncertainty, reliable optimum designs can be achieved through CDDO when the confidence level is set to 0.6, 0.7, 0.8, and 0.9. The reliabilities of proposed optimum designs increase with the level of target CL, which demonstrates that a higher confidence level can reduce the probability of overestimating the reliability. As shown in Fig. 10, a higher CL will push the optimum design into the safe region. From the results, we can conclude that the proposed approach is capable of providing a reliable optimum design when the surrogate model is constructed with limited data.

4.2 Case Study II: A Highly Nonlinear Example. This example has two constraints and two random design variables X_1 and X_2. It is assumed that all random variables are statistically independent and follow normal distributions with standard deviation 0.1. The CDDO problem is formulated as

$$\begin{aligned} &\text{minimize}: \text{Cost}(\mathbf{d}) = (d_1 - 3.7)^2 + (d_2 - 4)^2 \\ &\text{subject to}: \beta_C^i \geq \beta_t^i; \\ &\qquad \beta_C^i = \Phi_t^{-1}\left[\Pr\left(\hat{g}_{gp}^i(\mathbf{d})|_{\text{CL}} \geq 0\right)\right], i = 1, 2 \\ &\qquad \mathbf{d} = [X_1, X_2], \ \text{CL} = \text{CL}_t \\ &\qquad 0 \leq X_1 \leq 3.7, \ 0 \leq X_2 \leq 4 \\ &\text{where} \\ &\qquad G_1 = X_1 + X_2 - 3 \\ &\qquad G_2 = -X_1 \sin(4X_1) - 1.1X_2 \sin(2X_2) \end{aligned} \tag{30}$$

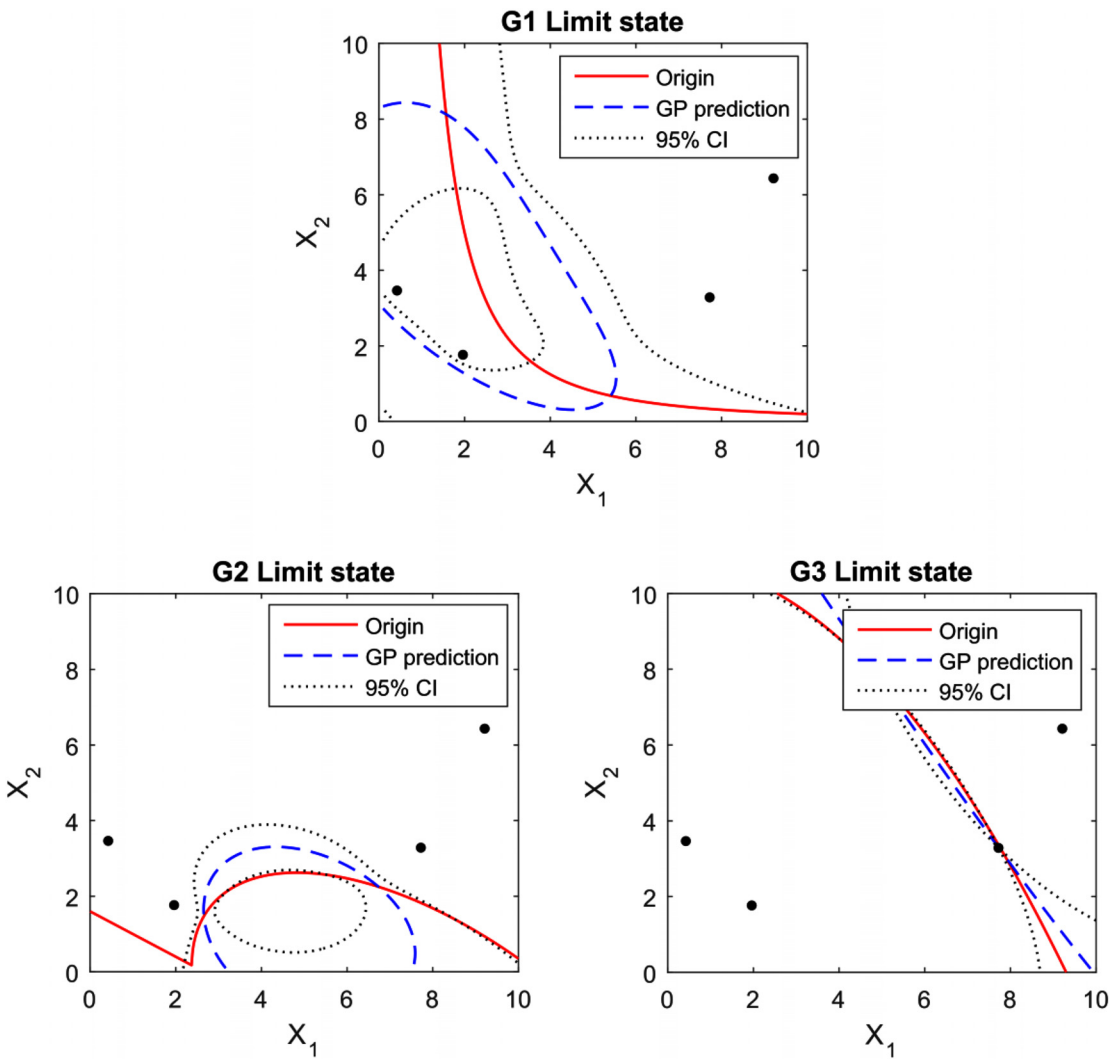


Fig. 7 Estimated versus origin LSFs (G_1, G_2, and G_3)

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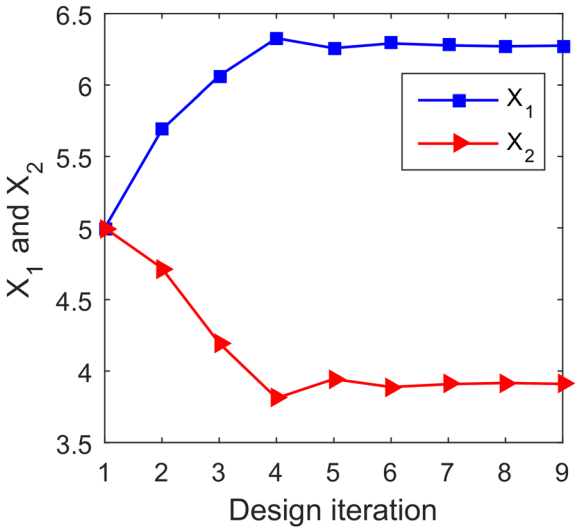


Fig. 8 Confidence-driven design optimization history of the two design variables

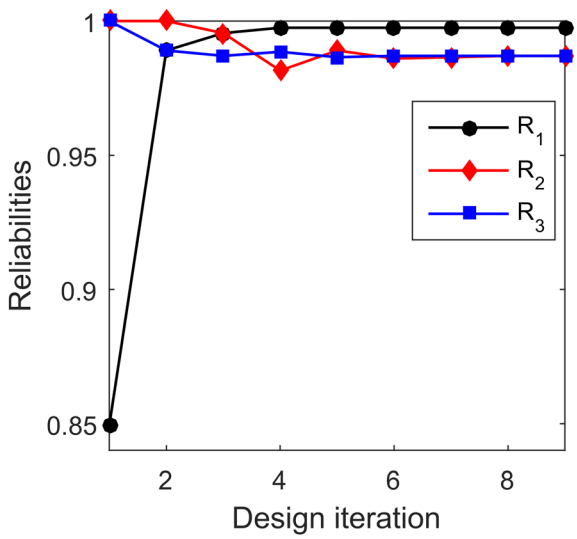


Fig. 9 Reliabilities estimation in each design iteration

The target reliability index for both constraints is set to $\beta_t = 2$ (the target reliability is approximated as 0.9772). To solve the design problem, the target confidence level is set to 0.7. Two GP models are constructed based on ten training sample generated by Latin hypercube sampling, corresponding to the performance function G_1 and G_2, respectively. In each design iteration, 10^5 MCS samples are used for reliability and sensitivity analysis. Starting from an initial design $\mathbf{d_0} = [2.5, 2.5]$, an optimum design $\mathbf{d_{opt}} = [2.9801, 2.7981]$ is obtained after ten iterations.

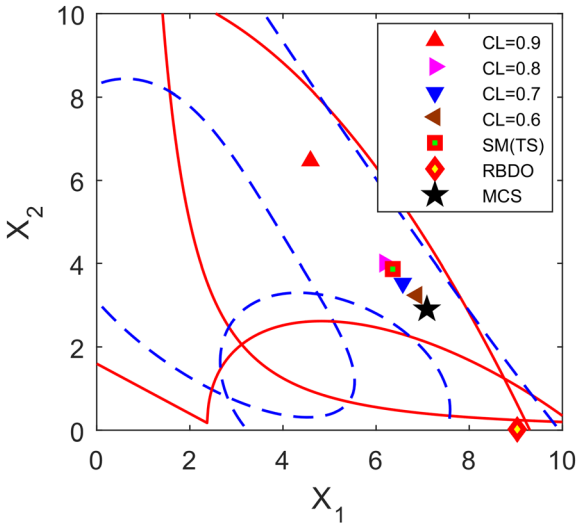


Fig. 10 Optimal designs obtained from different methods

The contour plot of estimated and origin limit state functions is depicted in Fig. 11, which shows the high nonlinearity of the performance function G_2. Using only 10 training samples is not sufficient for this two-dimensional problem due to the high nonlinearity. Table 4 summarizes the iterative history of design points, estimated reliabilities, and cost function values. The convergence of the estimated reliabilities and the design variables are shown in Figs. 12 and 13, respectively.

The comparison results are shown in Table 5, and Fig. 14 depicts the optimal designs obtained by different methods. It is obviously observed that conventional method failed to provide a reliable optimum design as the reliabilities of the performance function G2 0.3276 is less than the target reliability. Due to the limited data, the estimated safety margin using cross-validation approach is inaccurate, leading to an overestimated optimum design that has the largest cost function value. Though SM(TS) method offers an acceptable optimum at 90% conservative level, it requires 100 more functions evaluations than other approaches. By setting a target confidence level greater than 0.6, the proposed approach is capable of providing feasible optimal designs. Considering the tradeoff between the number of function evaluations and the cost function, we have tested another scenario that the GP models are constructed with 70 LHS samples. As the results indicated in Fig. 15, the optimal solution obtained through the proposed CDDO framework are very close to the accurate one from direct MCS. Due to the increased number of training data, the prediction variances become extremely small, and the confidence-based predictions did not change a lot when using different target confidence level.

4.3 Case Study III: Vehicle Side Impact. For vehicle side impact protection, the design must meet the safety regulations such as the national highway traffic safety administration and the European Enhances Vehicle-safety Committee (EEVC) side

Table 3 Comparison of optimal results for case study I

	Optimum	R_1	R_2	R_3	Cost	#F
MCS	[7.0891, 2.9153]	1	0.9850	0.9850	5.8262	3×10^6
RBDO	[9.0334, 0.0120]	0.2375	0.0045	0.8845	0.9786	3×6
SM(TS)	[6.3754, 3.8709]	1	1	0.9940	7.4955	3×56
SM(CV)	/	/	/	/	/	3×6
Proposed (CL = 0.6)	[6.8625, 3.2311]	1	0.9979	0.9880	6.3686	3×6
Proposed (CL = 0.7)	[6.5759, 3.5559]	1	0.9999	0.9925	6.9800	3×6
Proposed (CL = 0.8)	[6.1813, 4.0147]	1	1	0.9975	7.8334	3×6
Proposed (CL = 0.9)	[4.5946, 6.4883]	1	1	0.9991	11.894	3×6

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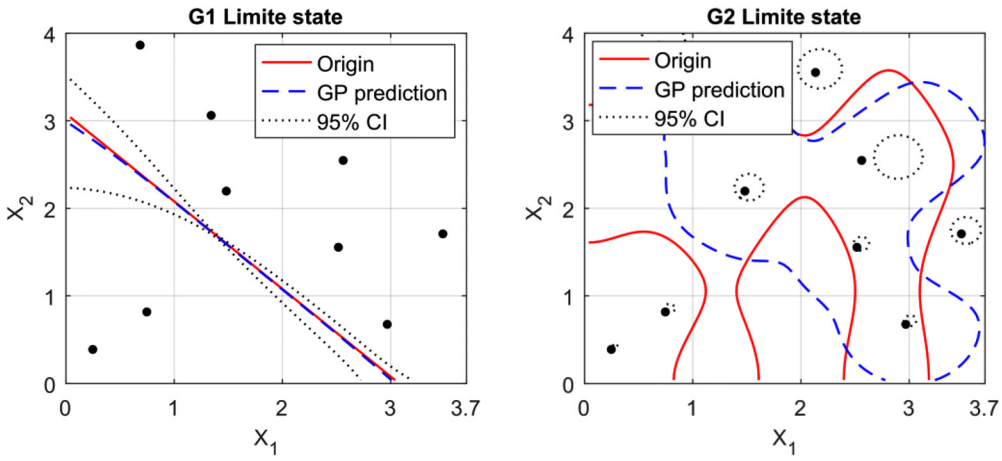


Fig. 11 Estimated versus origin LSFs (G_1 and G_2)

impact procedure. In this study, the EEVC test configuration [52] is employed to verify the dummy's responses of the abdomen load, rib deflections (upper, middle, and lower), viscous criterion, and pubic symphysis force. In addition, the velocity of the B-pillar at the middle point and the velocity of the front door at the B-pillar are two extra concerns in side impact analysis. Therefore, the vehicle side crash design case study involves a total number of nine design variables and two non-design random variables. It is assumed that all random variables are normally distributed, while the statistical information is given in Table 6. The two non-design random variables are barrier height (x_1) and hitting position (x_2) that follow a normal distribution with zero mean and standard deviation of 10.

In this design problem, ten constraint functions are considered for the car side impact, including the abdomen load (G_1), rib deflection (upper G_2, middle G_3, and lower G_4), viscous criterion (upper G_5, middle G_6, and lower G_7), pubic symphysis force (G_8), velocity of B-pillar at middle point (G_9), and velocity of front door at B-pillar (G_{10}). The detailed properties of the ten performance functions are given as

$$F_{\text{Abdom}} = 1.16 - 0.3717d_2d_4 - 0.00931d_2x_1 - 0.484d_3d_9 + 0.01343d_6x_1 \tag{31}$$

$$\text{deflection}_{\text{rib_u}} = 28.98 + 3.818d_3 - 4.2d_1d_2 + 0.0207d_5x_1 + 6.63d_6d_9 - 7.7d_7d_8 + 0.32d_9x_1 \tag{32}$$

$$\text{deflection}_{\text{rib_m}} = 33.86 + 2.95d_3 + 0.1792x_1 - 5.057d_1d_2 - 11d_2d_8 - 0.0215d_5x_1 - 9.98d_7d_8 + 22d_8d_9 \tag{33}$$

$$\text{deflection}_{\text{rib_l}} = 46.36 - 9.9d_2 - 12.9d_1d_8 + 0.1107d_3x_1 \tag{34}$$

Table 4 Design history for case study II

Iteration	Design point	$\hat{\mathbf{R}}_1$	$\hat{\mathbf{R}}_2$	Cost
1	[2.5000, 2.5000]	1.0000	0.8195	3.6900
2	[2.8335, 3.0035]	1.0000	0.6019	1.7439
3	[3.0461, 2.9240]	1.0000	0.7366	1.5853
4	[2.9826, 2.7341]	1.0000	0.9936	2.1171
5	[2.9582, 2.7644]	1.0000	0.9920	2.0770
6	[2.9720, 2.7908]	1.0000	0.9826	1.9921
7	[2.9751, 2.7941]	1.0000	0.9808	1.9797
8	[2.9754, 2.7940]	1.0000	0.9807	1.9794
9	[2.9799, 2.7979]	1.0000	0.9782	1.9637
10	[2.9801, 2.7981]	1.0000	0.9780	1.9629

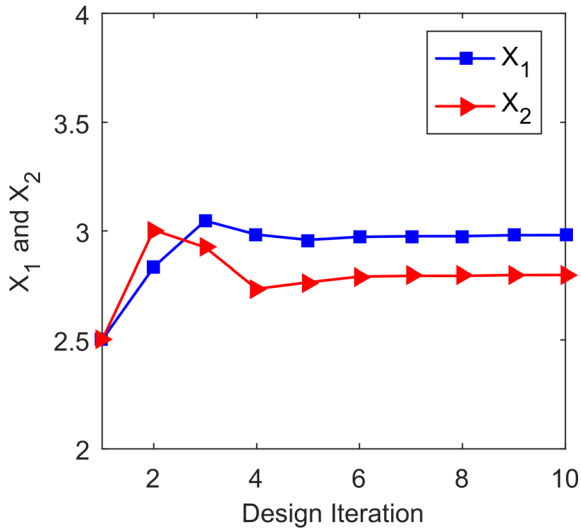


Fig. 12 Confidence-driven design optimization history of design variables

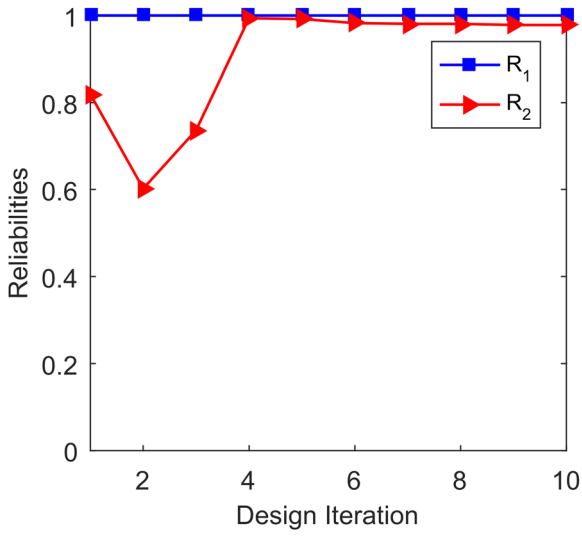


Fig. 13 Reliabilities estimation in each design iteration

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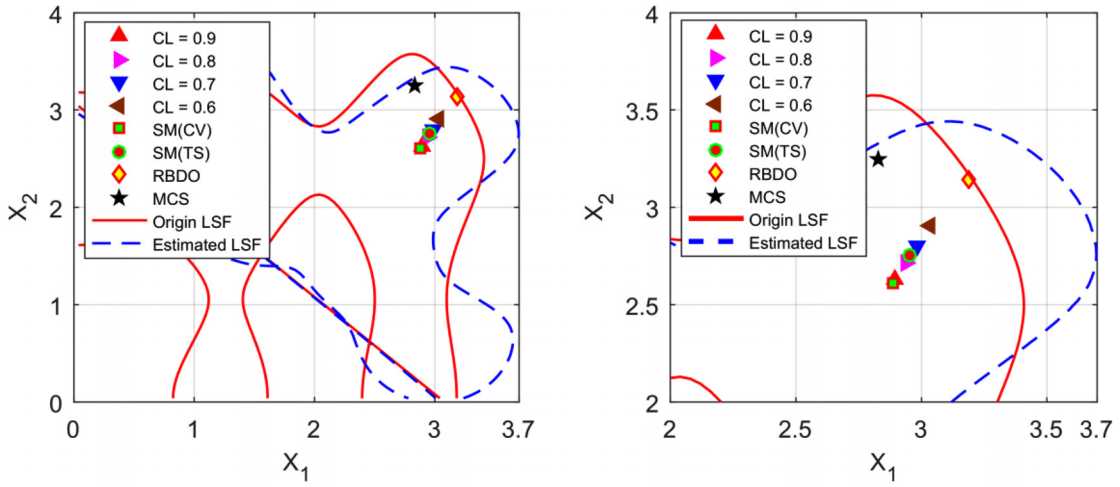


Fig. 14 Optimal designs obtained from different methods

Table 5 Comparison of optimal results for case study II

	Optimum	R_1	R_2	Cost	#F
MCS	[2.8276, 3.2493]	1	0.9772	1.3249	2×10^6
RBDO	[3.1889, 3.1442]	1	0.3276	0.9936	2×10
SM(TS)	[2.9533, 2.7506]	1	0.9997	2.1185	2×60
SM(CV)	[2.8839, 2.6090]	1	1	2.6009	2×10
Proposed (CL = 0.6)	[3.0370, 2.9053]	1	0.9759	1.6380	2×10
Proposed (CL = 0.7)	[2.9801, 2.7981]	1	0.9984	1.9629	2×10
Proposed (CL = 0.8)	[2.9384, 2.7153]	1	0.9998	2.2305	2×10
Proposed (CL = 0.9)	[2.8956, 2.6273]	1	1	2.5313	2×10

$$\begin{aligned} VC_{upper} = {} & 0.261 - 0.0159d_1d_2 - 0.188d_1d_8 - 0.019d_2d_7 \\ & + 0.0144d_3d_5 + 0.0008757d_5x_1 + 0.08045d_6d_9 \\ & + 0.00139d_8x_2 + 0.00001575x_1x_2 \end{aligned} \quad (35)$$

$$\begin{aligned} VC_{middle} = {} & 0.214 + 0.00817d_5 - 0.131d_1d_8 - 0.0704d_1d_9 \\ & + 0.03099d_2d_6 - 0.018d_2d_7 + 0.0208d_3d_8 \\ & + 0.121d_3d_9 - 0.00364d_5d_6 + 0.0007715d_5x_1 \\ & - 0.0005354d_6x_1 + 0.00121d_8x_2 + 0.00184d_9x_1 \\ & - 0.018d_2^2 \end{aligned} \quad (36)$$

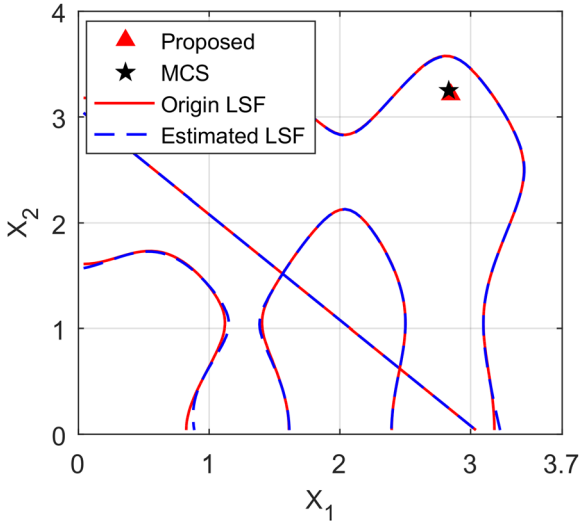


Fig. 15 Optimal design obtained with sufficient training data

Table 6 Properties of random variables for vehicle side impact

Design variables	Standard deviation	d_L	d_0	d_U
d_1(B-pillar inner)	0.03	0.5	1	1.5
d_2(B-pillar reinforce)	0.03	0.5	1	1.5
d_3(Floor side inner)	0.03	0.5	1	1.5
d_4(Cross member)	0.03	0.5	1	1.5
d_5(Door beam)	0.03	0.5	1	1.5
d_6(Door belt line)	0.03	0.5	1	1.5
d_7(Roof rail)	0.03	0.5	1	1.5
d_8(Mat. floor side inner)	0.006	0.192	0.3	0.345
d_9(Mat. floor side)	0.006	0.192	0.3	0.345
x_1(Barrier height)	10	/	0	/
x_2(Barrier hitting)	10	/	0	/

$$\begin{aligned} VC_{lower} = {} & 0.74 - 0.61d_2 - 0.163d_3d_8 + 0.001232d_3x_1 \\ & - 0.166d_7d_9 + 0.227d_2^2 \end{aligned} \quad (37)$$

$$\begin{aligned} Force_{public} = {} & 4.72 - 0.5d_4 - 0.19d_2d_3 - 0.0122d_4x_1 \\ & + 0.009325d_6x_1 + 0.000191x_2^2 \end{aligned} \quad (38)$$

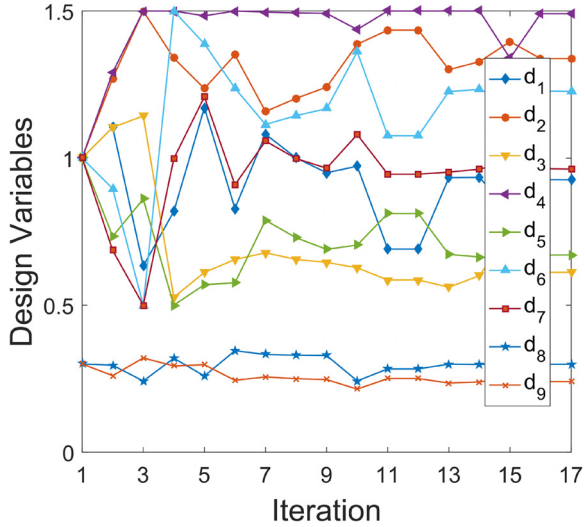


Fig. 16 Iterative CDDO history for design variables

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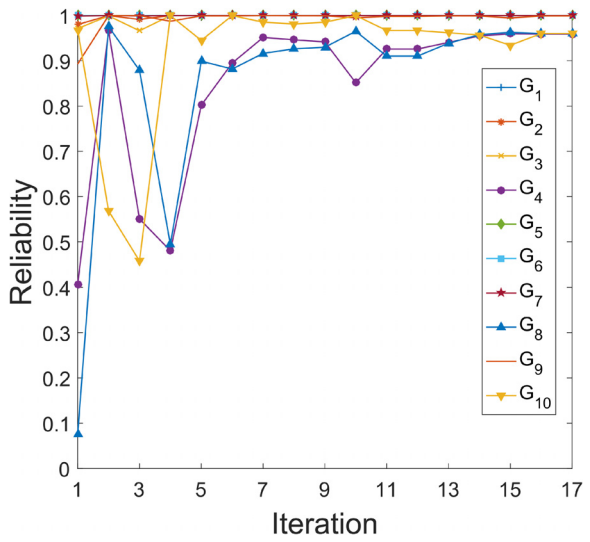


Fig. 17 Reliability history for ten probabilistic constraints

$$\text{velocity}_{B-\text{pillar}} = 10.58 - 0.674d_1d_2 - 1.95d_2d_8 + 0.02054d_3x_1 - 0.0198d_4x_1 + 0.028d_6x_1 \tag{39}$$

$$\text{velocity}_{\text{door}} = 16.45 - 0.489d_3d_7 - 0.843d_5d_6 + 0.0432d_9x_1 - 0.0556d_9x_2 - 0.000786x_2^2 \tag{40}$$

The objective of design optimization is to reduce the weight while maintaining the performance of the vehicle during a side impact test. Thus, the CDDO problem can be formulated as

$$\begin{aligned}
\text{minimize:}\quad & \text{Cost}(\mathbf{d}) \\
\text{subject to:}\quad & \beta_C^i \geq \beta_t^i; \\
& \beta_C^i = \Phi_t^{-1}\left[\Pr\left(\hat{g}_{gp}^i(\mathbf{d},\mathbf{x})|_{CL} \geq 0\right), i = 1 \sim 10\right] \\
& G_1 = 1 - F_{\text{Abdom}} \\
& G_{2/3/4} = 32 - \text{Deflection}_{\text{rib_u}}/\text{Deflection}_{\text{rib_m}}/\text{Deflection}_{\text{rib_l}} \\
& G_{5/6/7} = 0.32 - \text{VC}_{\text{upper}}/\text{VC}_{\text{middle}}/\text{VC}_{\text{lower}} \\
& G_8 = 4 - \text{Force}_{\text{public}} \\
& G_9 = 9.9 - \text{Velocity}_{B-\text{pillar}} \\
& G_{10} = 15.7 - \text{Velocity}_{\text{door}} \\
& \mathbf{d}_L \leq \mathbf{d} \leq \mathbf{d}_U, \text{CL} = \text{CL}_t
\end{aligned} \tag{41}$$

where objective function $Cost(\mathbf{d})$ represents the total weight, expressed as

$$\text{cost}(\mathbf{d}) = 1.98 + 4.90d_1 + 6.67d_2 + 6.98d_3 + 4.01d_4 + 1.78d_5 + 2.73d_7 \tag{42}$$

For the CDDO problem of car side impact, the target reliability is set to 0.96 ($\beta_t = 1.7507$) for all the ten constraints, and the target confidence level is set at 0.8. In this study, 200 training samples are first generated by Latin hypercube sampling and evaluated by the origin limit state functions, and then ten Gaussian process models are constructed according to the obtained training data sets. Following the procedure outlined in Subsec. 3.4, the CDDO starts with the initial design $\mathbf{d_0} = [1, 1, 1, 1, 1, 1, 1, 0.3, 0.3]$, and terminates at the optimum design $\mathbf{d_{opt}} = [0.9260, 1.3380, 0.6119, 1.4906, 0.6707, 1.2267, 0.9625, 0.2991, 0.2399]$ after 17 iterations. Figure 16 shows the iterative CDDO history of the 9 design variables, while the estimated reliabilities during design optimization are depicted in Fig. 17.

To demonstrate the efficacy of the proposed CDDO framework, conventional RBDO method, SM(TS) using 50 test samples, and

Table 7 Comparison of different optimal designs

	d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8	d_9	Cost
MCS	0.5023	1.3472	0.5015	1.3746	0.6260	1.4972	0.5021	0.3393	0.1932	24.9247
RBDO	0.7552	1.3334	0.5058	1.4805	0.5117	0.9948	0.7920	0.3067	0.2127	27.1154
SM(TS)	0.8774	1.4738	0.6028	1.5000	0.6497	1.2217	0.9460	0.2902	0.2331	30.0714
SM(CV)	0.9327	1.4427	0.6633	1.4399	0.7664	1.2489	0.9527	0.2955	0.2351	30.5422
CL = 0.6	0.8262	1.3288	0.5000	1.5000	0.5000	1.1423	0.9093	0.3035	0.2294	27.7686
CL = 0.7	0.8843	1.3294	0.5424	1.5000	0.6022	1.2042	0.9462	0.2995	0.2360	28.6361
CL = 0.8	0.9260	1.3380	0.6119	1.4906	0.6707	1.2267	0.9625	0.2991	0.2399	29.5119
CL = 0.9	0.9544	1.3604	0.7378	1.4151	0.7489	1.2259	0.9586	0.2980	0.2493	30.5044

Table 8 True reliabilities for the optimal designs

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	R_{10}
RBDO	1	1	1	0.9969	1	1	1	0.9191	1	0.3128
SM(TS)	1	1	1	1	1	1	1	0.9885	1	0.9624
SM(CV)	1	1	1	1	1	1	1	0.9879	1	0.9995
CL = 0.6	1	1	1	0.9986	1	1	1	0.9505	1	0.5080
CL = 0.7	1	1	1	0.9992	1	1	1	0.9698	1	0.8648
CL = 0.8	1	1	1	0.9992	1	1	1	0.9834	1	0.9763
CL = 0.9	1	1	1	0.9989	1	1	1	0.9876	1	0.9986

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SM(CV) method are employed to solve the same vehicle side crash design problem. As a reference, direct MCS method is employed to offer an optimum design for comparison. All the obtained optimum solutions are summarized in Table 7. To verify the accuracy of optimum designs obtained by different methods, the Monte Carlo simulation with 10^6 samples is employed to calculate the reliabilities for the ten limit state functions, and the results are shown in Table 8. It is observed that both SM(TS) and SM(CV) methods are capable of providing a reliable optimum design when the conservative level is set at 90%. However, SM(TS) method requires 500 more function evaluations to estimate the safety margin and SM(CV) method provides an optimum design with a higher cost than the proposed method. Without the consideration of surrogate model uncertainty, conventional RBDO method provides an optimum design that underestimates the probability of failure, as the reliabilities for G_8 and G_{10} are 0.9191 and 0.3128, which violates the probabilistic constraints. The optimum results obtained by CDDO at different target confidence levels are detailed in Table 7.

5 Conclusion

Although surrogate modeling techniques have been successfully employed for reliability-based design optimization, the performances of such methods for complex engineering applications are severely limited due to the expensive costs of collecting sufficient training data. When the training data is limited, most of the existing surrogate-based RBDO methods lack the capability to provide reliable optimum designs due to the ignorance of surrogate model uncertainty. Therefore, this paper presents a confidence-driven design optimization framework to achieve reliable optimum solutions for design problems with limited training data. To compensate the surrogate model uncertainty, a confidence-based Gaussian process model is developed to integrate with the MCS method for reliability assessment. By using the proposed approach, the type II classification errors that introduced by the surrogate model uncertainty can be compensated accordingly. As a result, the overestimation of reliability is avoided in the design optimization process. Furthermore, a new sensitivity analysis method is proposed to approximate the sensitivity of the transformed reliability index and thus facilitates the confidence-driven design optimization. The results demonstrate that the CDDO can efficiently handle surrogate model uncertainty and thus provide reliable optimum solutions by balancing the trade-off between the cost and the risk of overestimating the reliability.

Nomenclature

CL = confidence level
CL_t = target confidence level
$\mathrm{Cost}(.)$ = cost function
$\mathbf{d}$ = design variables
$\mathbf{D}$ = training data set
$f_x(.)$ = probability density function
$\mathbf{G}$ = training responses
$G(.)$ = performance function
$\hat{\mathrm{g}}_{\mathrm{gp}}(.)$ = Gaussian process model
$\hat{\mathrm{g}}_{\mathrm{gp}}(.)|_{\mathrm{CL}}$ = confidence-based prediction at given CL
P_f = probability of failure
R_t = target reliability
$\hat{R}_{\mathrm{CL}}$ = estimated reliability at a given confidence level
$\mathbf{x}$ = random variables
$\mathbf{X}$ = training samples
$\mathbf{x_r}$ = one random realization of input $\mathbf{x}$
$\mathbf{X_m}$ = Monte Carlo samples of input $\mathbf{x}$
β_c = transformed reliability index
$\mu_{\mathrm{gp}}(.)$ = prediction mean from GP model
μ_{C} = mean of confidence-based predictions
$v_{\mathrm{gp}}(.)$ = prediction variance from GP model
v_{C} = variance of confidence-based predictions
$\Phi(\cdot)$ = standard normal cumulative distribution function
$\Phi^{-1}{}_{\mathrm{t}}(\cdot)$ = inverse CDF of truncated normal distribution

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